

# PISA 2025 Insight Report: Mongolia and Rwanda

This report compares PISA 2025<sup>1</sup> performance in Mongolia and Rwanda with a pooled OECD comparison group. Mathematics is the primary outcome, with science and reading providing context. The analysis examines socioeconomic status (ESCS), gender, grade repetition, school location, and material and staff shortages. Achievement estimates use student weights and all 10 plausible values. Findings are descriptive rather than causal; missing-data patterns and a school-location anomaly limit some comparisons.

› 1. Package Imports

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› 2. Downloading PISA Files

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∨ 3. Data Loading and Column Selection

› 3A. Load School Data

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› 3B. Select and Load Student Data

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∨ 4. Column Standardization

› 4A. Standardize School Column Names

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› 4B. Standardize Student Column Names

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∨ 5. Data Cleaning & Missingness Analysis

› 5A. Define Missing-Value Codes

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› 5B. Map Missing Codes to Variables

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› 5C. Create Analysis Groups

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› 5D. Clean School Data and Check Missingness

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› 5E. School Missingness Visualization

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› 5F. Clean Student Data and Check Missingness

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› 5G. Student Missingness Visualization

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5H. Missingness and Mathematics Performance

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|   | Analysis Group | Variable            | Missing N | Observed N | Missing Math | Observed Math | Difference |  |
|---|----------------|---------------------|-----------|------------|--------------|---------------|------------|--|
| 0 | Mongolia       | socioeconomic_index | 245       | 6901       | 351.16       | 421.16        | -70.01     |  |
| 1 | Mongolia       | grade_repetition    | 36        | 7110       | 364.84       | 418.94        | -54.09     |  |
| 2 | Rwanda         | socioeconomic_index | 442       | 6180       | 267.13       | 314.94        | -47.81     |  |
| 3 | Rwanda         | grade_repetition    | 335       | 6287       | 273.89       | 313.70        | -39.81     |  |
| 4 | OECD           | socioeconomic_index | 30369     | 284945     | 438.02       | 462.27        | -24.25     |  |
| 5 | OECD           | grade_repetition    | 30025     | 285289     | 475.41       | 454.04        | 21.38      |  |

Analysis Note: Data Missingness and Integrity

ESCS missingness is **3.43% in Mongolia**, **6.67% in Rwanda**, and **9.63% in the pooled OECD comparison group**. Repetition missingness is **0.50%**, **5.06%**, and **9.52%**, respectively. Weights are complete; achievement and gender are virtually complete, with one OECD student missing all 30 plausible values and 15 missing gender. OECD school-level missingness is roughly **12–14%** for location and shortage measures. School-linked OECD analyses exclude Iceland and Luxembourg because school IDs are unavailable.

Students missing ESCS score about **70 points lower in Mongolia**, **48 in Rwanda**, and **24 in the pooled OECD comparison group**. Missing repetition also accompanies lower mathematics performance in Mongolia and Rwanda. The pooled OECD repetition pattern reverses the within-country pattern because country availability differs. ESCS and repetition analyses therefore describe selected observed samples. School-condition missingness has heterogeneous achievement associations.

6. Overall PISA Performance

For each subject and analysis group, we calculate the student-weighted mean separately for each of PISA's 10 plausible values and then average those 10 estimates.<sup>2</sup> The plausible values are alternative estimates of achievement, not 10 separate tests.

6A. Weighted Plausible Value Function

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6B. Overall PISA Performance Summary

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| === PISA 2025 Weighted Performance Summary === |             |         |         |
|------------------------------------------------|-------------|---------|---------|
|                                                | Mathematics | Reading | Science |
| Rwanda                                         | 311.65      | 300.94  | 317.29  |
| Mongolia                                       | 418.64      | 374.25  | 424.03  |
| OECD                                           | 458.37      | 467.21  | 484.46  |

Analysis Note: Overall Performance

Mongolia outperforms Rwanda across all three domains: the mathematics gap is **106.99 points**, science **106.74**, and reading **73.31**. Both countries remain below the pooled OECD comparison group; Mongolia's mathematics shortfall is **39.73 points**. Its reading and science shortfalls are **92.96** and **60.43 points**, respectively. Rwanda's shortfalls are larger. These results describe substantial performance differences without explaining their sources. The remaining analysis examines associations with student and school characteristics.

7. Comparison of Student and School Characteristics

7A. Weighted Summary Functions

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7B. Student and School Characteristics Summary

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|                | Male Students (%) | Grade Repetition Rate (%) | Mean ESCS (Socioeconomic) | Reported Urban School Location (%) | Educational Material Shortage (Mean) | Staff Shortage Index (Mean) |  |
|----------------|-------------------|---------------------------|---------------------------|------------------------------------|--------------------------------------|-----------------------------|--|
| Analysis Group |                   |                           |                           |                                    |                                      |                             |  |
| Mongolia       | 49.12             | 5.65                      | -0.78                     | 47.06                              | 0.04                                 | 0.06                        |  |
| Rwanda         | 43.16             | 55.70                     | -2.15                     | 14.41                              | 0.67                                 | 0.26                        |  |
| OECD           | 49.43             | 11.18                     | -0.19                     | 48.94                              | -0.07                                | 0.28                        |  |

**Location note.** Eleven Rwandan schools, linked to **346 students**, have raw location code 6, labeled communities above 10 million people. Eight schools are in Kigali and one each in South, North, and West; even Kigali is below that threshold.<sup>3</sup> All eleven have urban sampling

strata, supporting broad urban grouping rather than literal megacity interpretation. The original cause remains unresolved; these observations are retained and flagged. The public file cannot distinguish unusual responses from a country-specific convention or an earlier coding transformation.

Analysis Note: Student and School Characteristics

Grade repetition is **55.70% in Rwanda**, compared with **5.65% in Mongolia** and **11.18% in the OECD comparison group**. Mean ESCS is substantially lower in Rwanda (**-2.15**) than Mongolia (**-0.78**) or the OECD comparison group (**-0.19**).

Among students with observed school-location data, **14.41%** of Rwanda's represented students attend schools reported in the broad urban categories, compared with **47.06% in Mongolia** and **48.94% in the pooled OECD comparison group**. Rwanda reports greater educational material shortages (**0.67**) than Mongolia (**0.04**) or the OECD comparison group (**-0.07**). Staff-shortage means are **0.26**, **0.06**, and **0.28**, respectively.

8. Association with Mathematics performance

8A. Merge Student and School Data

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8B. Weighted Mathematics Performance by Category

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8C. Gender and Grade Repetition

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=== Mathematics Performance by Gender and Grade Repetition ===

|                | Female/Other Math | Male Math | Gender Gap (Male - Female/Other) | No Repetition | Repeated Grade | Repetition Gap (Repeated - No Repetition) |
|----------------|-------------------|-----------|----------------------------------|---------------|----------------|-------------------------------------------|
| Analysis Group |                   |           |                                  |               |                |                                           |
| Mongolia       | 422.75            | 414.38    | -8.37                            | 424.08        | 332.92         | -91.17                                    |
| Rwanda         | 301.73            | 324.70    | 22.97                            | 339.25        | 293.39         | -45.87                                    |
| OECD           | 449.13            | 467.82    | 18.70                            | 463.56        | 378.37         | -85.19                                    |

Analysis Note: Gender and Grade Repetition

Male students score about **23 points higher** than the female/other group in Rwanda and **19 points higher** in the OECD comparison group, while Mongolia shows an **8-point** difference in the opposite direction. Grade repetition shows much larger differences: repeaters score about **91 points lower in Mongolia**, **46 lower in Rwanda**, and **85 lower in the OECD comparison group**. These differences are descriptive, not causal.

8D. Mathematics Performance Across ESCS Ranges

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=== Mathematics Performance Across Common ESCS Ranges ===

|              | Mongolia | Rwanda | OECD   |
|--------------|----------|--------|--------|
| Below -2.0   | 373.89   | 308.22 | 381.73 |
| -2.0 to -1.0 | 400.64   | 311.44 | 414.60 |
| -1.0 to 0.0  | 423.18   | 328.64 | 449.65 |
| 0.0 to 1.0   | 461.94   | 356.10 | 490.60 |
| Above 1.0    | 484.69   | 348.51 | 516.69 |

=== Student Counts Across Common ESCS Ranges ===

| escs_range     | Below -2.0 | -2.0 to -1.0 | -1.0 to 0.0 | 0.0 to 1.0 | Above 1.0 |
|----------------|------------|--------------|-------------|------------|-----------|
| analysis_group |            |              |             |            |           |
| Mongolia       | 782        | 2057         | 2119        | 1806       | 137       |
| Rwanda         | 3536       | 1361         | 868         | 379        | 36        |
| OECD           | 9265       | 37408        | 79612       | 109784     | 48876     |

Next steps: [Generate code with escs\\_range\\_counts](#) [New interactive sheet](#)

Analysis Note: Socioeconomic Status

Mathematics generally increases with ESCS. Mongolia rises from **373.89** below ESCS -2 to **484.69** above 1, while the OECD comparison group rises from **381.73** to **516.69**. Rwanda rises from **308.22** below -2 to **356.10** between 0 and 1, then falls slightly to **348.51** above 1; only **36 Rwandan students** are in that highest category. Mongolia remains above Rwanda within every common ESCS range.

8E. Educational Material Shortages

[Show code](#)

| === Mathematics Performance by Educational Material Shortage === |          |        |        |  |
|------------------------------------------------------------------|----------|--------|--------|--|
|                                                                  | Mongolia | Rwanda | OECD   |  |
| Low                                                              | 429.19   | 334.59 | 467.50 |  |
| Moderate                                                         | 418.56   | 312.45 | 458.08 |  |
| High                                                             | 406.53   | 304.38 | 444.31 |  |

Analysis Note: Educational Material Shortages

Mathematics performance falls as reported educational material shortages increase in all three groups. Students in the high-shortage category score about **23 points lower** than those in the low-shortage category in Mongolia, **30 points lower in Rwanda**, and **23 points lower in the OECD comparison group**.

8F. Mathematics Performance by School Location

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| === Mathematics Performance by Reported School Location === |          |        |        |  |
|-------------------------------------------------------------|----------|--------|--------|--|
|                                                             | Mongolia | Rwanda | OECD   |  |
| Rural (<3,000)                                              | 376.00   | 304.00 | 410.15 |  |
| Town (3,000-100,000)                                        | 401.98   | 315.96 | 458.78 |  |
| Urban (>100,000)                                            | 438.09   | 344.77 | 464.70 |  |

[Show code](#)

| === Rwanda School Location Sensitivity Check === |                    |            |                 |                        |
|--------------------------------------------------|--------------------|------------|-----------------|------------------------|
|                                                  | Urban Students (%) | Urban Math | Rural/Town Math | Urban - Rural/Town Gap |
| Scenario                                         |                    |            |                 |                        |
| All Reported Locations                           | 14.41              | 344.77     | 306.43          | 38.34                  |
| Excluding Category 6                             | 9.86               | 342.38     | 306.43          | 35.95                  |

Analysis Note: School Location

Rwandan students in reported urban schools score higher than those in rural or town schools. Excluding the 11 schools with anomalous location responses changes the urban–rural/town gap from about **38 to 36 points**, preserving its descriptive direction. This sensitivity check does not validate detailed population categories, which remain uncertain.

8G. Weighted Correlation Analysis

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| === Weighted Correlations with Mathematics Performance === |       |                   |                |  |
|------------------------------------------------------------|-------|-------------------|----------------|--|
|                                                            | ESCS  | Material Shortage | Staff Shortage |  |
| Analysis Group                                             |       |                   |                |  |
| Mongolia                                                   | 0.376 | -0.124            | -0.017         |  |
| Rwanda                                                     | 0.143 | -0.161            | -0.098         |  |
| OECD                                                       | 0.395 | -0.103            | -0.049         |  |

Analysis Note: Correlations with Mathematics

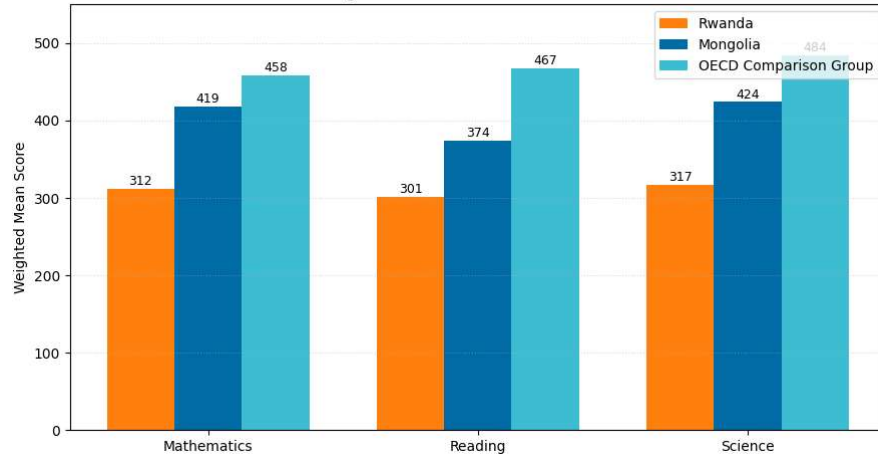
ESCS has the clearest positive correlation with mathematics in Mongolia (**0.376**) and the OECD comparison group (**0.395**), while the relationship is weaker in Rwanda (**0.143**). Material shortages are negatively correlated with mathematics in all three groups, with the largest absolute correlation in Rwanda (**–0.161**). Staff-shortage correlations are weaker. These simple correlations do not account for other differences between students or schools.

9. Key Visualizations

9A. Overall PISA Performance

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Figure 1: Overall PISA Performance



## 9B. Student Population Characteristics

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Figure 2A: Socioeconomic Status

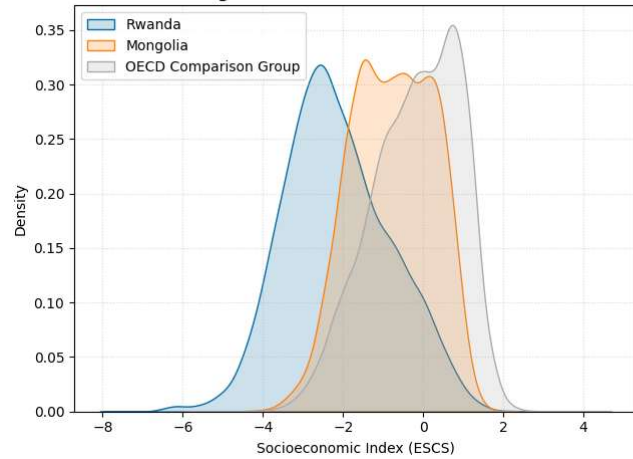
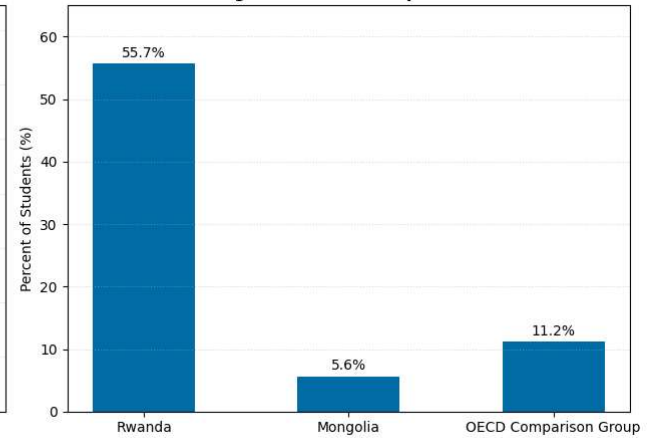


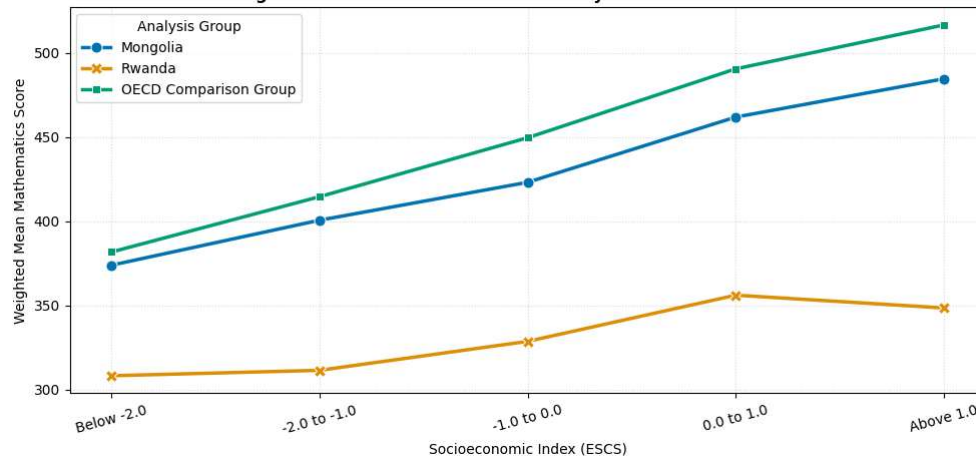
Figure 2B: Grade Repetition



## 9C. Mathematics Across ESCS Ranges

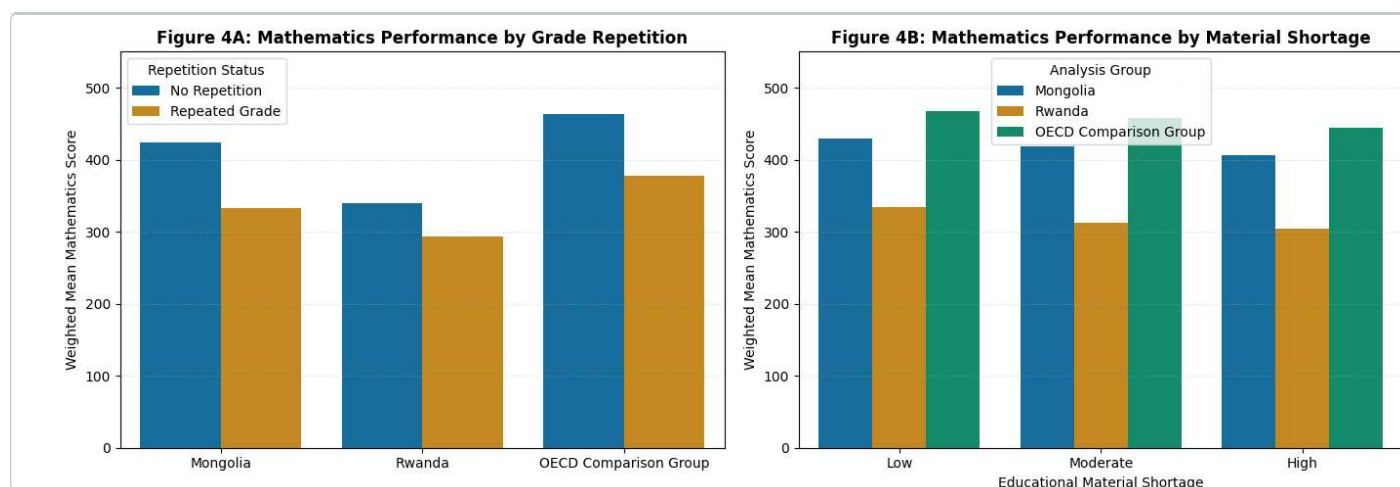
Show code

Figure 3: Mathematics Performance by Socioeconomic Status



## 9D. Grade Repetition and Material Shortage Comparisons

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## 10. Conclusion and Recommendations

Mongolia substantially outperforms Rwanda in PISA 2025, and the within-country results highlight socioeconomic status, grade repetition, and material shortages as important areas for further study. The analysis also shows that data quality affects interpretation: missing ESCS and repetition information is associated with student performance, and Rwanda's detailed school-location responses contain an implausible category. Future data collection could flag implausible responses during collection, monitor missing responses before fieldwork ends, and cross-check school information against administrative or sampling records. These findings should guide further policy evaluation rather than be treated as evidence that changing any single factor would cause higher mathematics achievement.

## Endnotes

1. OECD, *PISA 2025 Database*, <https://www.oecd.org/en/data/datasets/pisa-2025-database.html>
2. OECD, *PISA Research Documentation: How to Prepare and Analyse the PISA Database*, <https://www.oecd.org/en/about/programmes/pisa/how-to-prepare-and-analyse-the-pisa-database.html>
3. National Institute of Statistics of Rwanda, *Fifth Rwanda Population and Housing Census 2022: Main Indicators Report*, [https://alpha.statistics.gov.rw/sites/default/files/documents/2025-02/RPHC5\\_MainIndicatorsReport\\_Final.pdf](https://alpha.statistics.gov.rw/sites/default/files/documents/2025-02/RPHC5_MainIndicatorsReport_Final.pdf)

## Generative AI Acknowledgment

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